

AYAKA YORIIHIRO

ayaka@cs.cornell.edu <https://ayakayorihiro.github.io/>

EDUCATION

Cornell University

Ph.D. in Computer Science.

M.S. in Computer Science.

Advisor: Adrian Sampson.

September 2020 - May 2027

Swarthmore College

B.A., Computer Science and Music (double major).

Elected Phi Beta Kappa.

Advisor: Zachary Palmer.

September 2016 - May 2020

HONORS AND AWARDS

Qualcomm Innovation Fellowship Finalist

2026

NSF GRFP Honorable Mention

2022

Cornell University Outstanding Teaching Award

2022

Cornell University Fellowship

2020

Lockwood Fellowship, Swarthmore College

2020

PUBLICATIONS

Understanding Accelerator Compilers via Performance Profiling *OOPSLA 2026 (to appear)*

Ayaka Yorihiro, Griffin Berstein, Pedro Pontes García, Kevin Laeuffer, and Adrian Sampson.

ACM SIGPLAN Conference on Object-oriented Programming, Systems, Languages, and Applications (OOPSLA).

A FIRRTL Backend for the Calyx High-Level Accelerator

OSDA 2024

Compilation Infrastructure

Ayaka Yorihiro, Griffin Berstein, Kevin Laeuffer, and Adrian Sampson.

Workshop on Open-Source Design Automation (OSDA).

eMOP: A Maven Plugin for Evolution-Aware Runtime Verification

RV 2023

Ayaka Yorihiro, Pengyue Jiang, Valeria Marqués, Benjamin Carleton, and Owolabi Legunsen.

International Conference on Runtime Verification (RV).

A Set-Based Context Model for Program Analysis

APLAS 2020

Leandro Fachinetti, Zachary Palmer, Scott F. Smith, Ke Wu, Ayaka Yorihiro.

Asian Symposium on Programming Languages and Systems (APLAS).

TEACHING EXPERIENCE

Co-instructor for CS 3110: Data Structures and Functional Programming

Spring 2026

Cornell University, Ithaca, NY.

Instructor of Record for CS 2110: Object-Oriented Programming and Data Structures

Cornell University, Ithaca, NY.

Summer 2024

Teaching Assistant for CS 2110: Object-Oriented Programming and Data Structures

Cornell University, Ithaca, NY.

Fall 2023, Spring 2024

Teaching Assistant for CS 5154: Software Testing*Fall 2021, Fall 2022*

Cornell University, Ithaca, NY.

Teaching Assistant for CS 73: Programming Languages*Spring 2020*

Swarthmore College, Swarthmore, PA.

Teaching Assistant for CS 46: Theory of Computation*Spring 2020*

Swarthmore College, Swarthmore, PA.

Student Mentor for CS 21: Introduction to Computer Science*Fall 2018*

Swarthmore College, Swarthmore, PA.

PROFESSIONAL EXPERIENCE**Research Assistant***Summer 2021, Jan - Aug 2022, Spring 2023, Sept 2024 - Present*

Cornell University, Ithaca, NY.

Research Assistant*Jan - Dec 2019*

Swarthmore College, Swarthmore, PA.

C++ Engineering Intern*Summer 2018*

Electronic Arts, Orlando, FL.

PRESENTATIONS**Languages and Tools for Hardware Accelerator Design***November 2024*

Comp 321: Programming Languages Guest Lecture

Ithaca College, Ithaca, NY.

Languages and Tools for Hardware Accelerator Design*September 2024*

Swarthmore College, Swarthmore, PA.

Towards Better Specifications for Runtime Verification*Fall 2021*

Programming Languages Discussion Group Retreat

Cornell University, Ithaca, NY.

SERVICE**Artifact Evaluation Committee***TACAS 2026*

International Conference on Tools and Algorithms for the Construction and Analysis of Systems.

Department of Computer Science, PL Website Czar*Spring 2025 - Present*

Cornell University, Ithaca, NY.

Department of Computer Science, Ph.D. Mentorship Czar*Spring 2022 - Present*

Cornell University, Ithaca, NY.

Co-organizer, Joint UT-Cornell Software Engineering Seminar*Summer 2022***CS Ph.D. Student Visit Day Student Panelist***Spring 2022*

Cornell University, Ithaca, NY.

CS Ph.D. Student Visit Day Organizer*Spring 2021*

Cornell University, Ithaca, NY.

SKILLS AND INTERESTS**Programming Languages**Rust, Java, Python, Bash, OCaml, C/C++, L^AT_EX**Languages**

English (Fluent), Japanese (Fluent)

Interests and Activities

Viola, Orchestra, Chamber Music, Special Needs